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Peter Hall

The Bootstrap and Edgeworth Expansion



Springer Science+Business Media, LLC

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Mathematical Subject Classification: 62E20, 62G09, 62E25

With six figures.

ISBN 978-0-387-94508-8

ISBN 978-1-4612-4384-7 (eBook)

DOI 10.1007/978-1-4612-4384-7

Library of Congress Cataloging-in-Publication Data
Hall, P.

The bootstrap and Edgeworth expansion / Peter Hall.

p. cm. — (Springer series in statistics)

Includes bibliographical references and index.

ISBN 0-387-94508-3.

1. Bootstrap (Statistics) 2. Edgeworth expansions. I. Title.

II. Series.

QA276.8.H34 1992

519.5—dc20

91-34951

© 1992 Springer Science+Business Media New York

Originally published by Springer-Verlag New York

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Production managed by Hal Henglein; manufacturing supervised by Jacqui Ashri.

Preface

This monograph addresses two quite different topics in the belief that each can shed light on the other. First, we lay the foundation for a particular view, even a personal view, of theory for Brad Efron's bootstrap. This topic is of course relatively new, its history extending over little more than a decade. Second, we give an account of theory for Edgeworth expansion, which has recently received a revival of interest owing to its usefulness for exploring properties of contemporary statistical methods. The theory of Edgeworth expansion is now a century old, if we date it from Chebyshev's paper in 1890.

Methods based on Edgeworth expansion can help explain the performance of bootstrap methods, and on the other hand, the bootstrap provides strong motivation for reexamining the theory of Edgeworth expansion. We do not claim to provide a definitive account of the totality of either subject. This is particularly so in the case of Edgeworth expansion, where our description of the theory is almost entirely restricted to results that are useful in exploring properties of the bootstrap. On the part of the bootstrap, our theoretical study is largely confined to a collection of problems that may be treated using classical theory for sums of independent random variables — the so-called “smooth function model.” The corresponding class of statistics is wide (it includes sample means, variances, correlations, ratios and differences of means and variances, etc.) but is by no means comprehensive. It does not include, for example, the sample median or other sample quantiles, whose treatment is relegated to an appendix. Our study of the bootstrap is directed mainly at problems involving distribution estimation, since the theory of Edgeworth expansions is most relevant there. (However, our introduction to bootstrap methods in Chapter 1 is more wide-ranging.) We consider linear regression and nonparametric regression, but not general linear models or functional regression. We treat some but by no means all possible approaches to using the bootstrap in density estimation. We do not treat bootstrap methods for dependent data. There are many other examples of our selectivity, which is designed to present a coherent and unified account of theory for the bootstrap.

Chapter 1 is about the bootstrap, with almost no mention of Edgeworth expansion; Chapter 2 is about Edgeworth expansion, with scarcely a word about the bootstrap; and Chapter 3 brings these two themes together, using Edgeworth expansion to explore and develop properties of the bootstrap. Chapter 4 continues this approach along less conventional lines, and Chapter 5 sketches details of mathematical rigour that are missing from earlier chapters. Finally, five appendices present material that

complements the main theoretical themes of the monograph. For example, Appendix II on Monte Carlo simulation summarizes theory behind the numerical techniques that are essential to practical implementation of the bootstrap methods described in Chapters 1, 3, and 4. Each chapter concludes with a bibliographical note; citations are kept to a minimum elsewhere in the chapters.

I am grateful to D.R. Cox, D.J. Daley, A.C. Davison, B. Efron, N.I. Fisher, W. Härdle, P. Kabaila, J.S. Marron, M.A. Martin, Y.E. Pittelkow, T.P. Speed and T.E. Wehrly for helpful comments and corrections of some errors. J., N. and S. provided much-needed moral support throughout the project. Miss N. Chin typed all drafts with extraordinary accuracy and promptness, not to say good cheer. My thanks go particularly to her. Both of us are grateful to Graeme Bailey for his help, at all hours of the day and night, in getting the final $\text{\TeX}$ manuscript to the production phase.

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Common Notation

Alphabetical Notation

He_j : j th Hermite polynomial; the functions He_j , $j \geq 0$, are orthogonal with respect to the weight function $e^{-x^2/2}$. (See Section 2.2.)

$\widehat{I}$: percentile bootstrap confidence interval.

$\widehat{J}$: percentile- t bootstrap confidence interval.

I, J : nonbootstrap version of $\widehat{I}, \widehat{J}$, obtained by reference to the distributions of $(\hat{\theta} - \theta_0)/\sigma$, $(\hat{\theta} - \theta_0)/\hat{\sigma}$ respectively.

$\mathcal{I}, \mathcal{J}$: generic confidence intervals.

(If numeric subscripts are appended to $\widehat{I}, \widehat{J}, I, J, \mathcal{I}$, or $\mathcal{J}$, then the first subscript (1 or 2) indicates the number of sides to the interval.)

$I(\cdot)$: indicator function.

n : sample size.

p, q (denoting functions; with or without subscripts) : polynomials in Edgeworth or Cornish-Fisher expansions. We always reserve q for the case of a Studentized statistic.

$\mathbb{R}$: real line.

$\mathbb{R}^d$: d -dimensional Euclidean space.

$\mathcal{R}$: subset of $\mathbb{R}^d$.

t_α : α -level quantile of Student's t distribution with $n - \nu$ degrees of freedom, for a specified integer $\nu \geq 1$.

u_α, v_α : α -level quantiles of distributions of $n^{1/2}(\hat{\theta} - \theta_0)/\sigma$, $n^{1/2}(\hat{\theta} - \theta_0)/\hat{\sigma}$ respectively. That is, u_α and v_α solve the equations

$$P\{n^{1/2}(\hat{\theta} - \theta_0)/\sigma \leq u_\alpha\} = P\{n^{1/2}(\hat{\theta} - \theta_0)/\hat{\sigma} \leq v_\alpha\} = \alpha.$$

$\hat{u}_\alpha, \hat{v}_\alpha$: bootstrap estimates of u_α, v_α , respectively.

$\mathcal{X} = \{X_1, \dots, X_n\}$: random sample from population. (A generic X_i is denoted by X .)

$\mathcal{X}^* = \{X_1^*, \dots, X_n^*\}$: resample obtained by sampling from $\mathcal{X}$ with replacement. (A generic X_i^* is denoted by X^* .)

$\bar{X}, \bar{X}^*$: means of $\mathcal{X}, \mathcal{X}^*$ respectively.

z_α : α -level quantile of Standard Normal distribution; that is, $z_\alpha = \Phi^{-1}(\alpha)$.

α : nominal coverage level of confidence interval; $\xi = \frac{1}{2}(1 + \alpha)$.

γ : skewness of a distribution. (Generally, $\gamma = \kappa_3$.)

θ_0 : true value of unknown parameter θ .

$\hat{\theta}$: estimate of θ , computed from $\mathcal{X}$.

$\hat{\theta}^*$: bootstrap version of $\hat{\theta}$, computed from $\mathcal{X}^*$. (If bootstrap quantities are computed by Monte Carlo simulation, involving calculation of a sequence $\hat{\theta}_1^*, \dots, \hat{\theta}_B^*$, then $\hat{\theta}^*$ denotes a generic $\hat{\theta}_b^*$.)

κ_j : j th cumulant of a distribution.

κ (without subscript) : kurtosis of a distribution. (Generally, $\kappa = \kappa_4$.)

μ : population mean.

ξ : see α .

σ^2 : asymptotic variance of $n^{1/2} \hat{\theta}$.

$\hat{\sigma}^2$: estimate of σ^2 , computed from $\mathcal{X}$.

$\hat{\sigma}^{*2}$: bootstrap version of $\hat{\sigma}^2$, computed from $\mathcal{X}^*$.

ϕ, Φ (without subscripts) : Standard Normal density, distribution functions respectively.

χ : characteristic function.

χ^* : bootstrap version of χ .

Nonalphabetical Notation, and Notes

The transpose of a vector or matrix $\mathbf{v}$ is indicated by $\mathbf{v}^T$. If $\mathbf{v}$ is a vector, its i th element is denoted by $v^{(i)} = (\mathbf{v})^{(i)}$. If $\mathbf{x}, \mathbf{y}$ are two vectors of length d , then $\mathbf{x} \geq \mathbf{y}$ means $x^{(i)} \geq y^{(i)}$ for $1 \leq i \leq d$.

Complementation is denoted by $\sim$. For example, if $\mathcal{R} \subseteq \mathbb{R}^d$ then

$$\tilde{\mathcal{R}} = \mathbb{R}^d \setminus \mathcal{R} = \{\mathbf{x} \in \mathbb{R}^d : \mathbf{x} \notin \mathcal{R}\}.$$

Given a sample $\mathcal{X} = \{X_1, \dots, X_n\}$, the empirical measure is the measure that assigns to a set $\mathcal{R}$ a value equal to the proportion of sample values contained within $\mathcal{R}$. The empirical distribution function (d.f.) and empirical characteristic function (ch.f.) are respectively the d.f. and ch.f. of the empirical measure.

If $\{\Delta_n\}$ is a sequence of random variables and $\{\delta_n\}$ is a sequence of positive constants, we write $\Delta_n = O_p(\delta_n)$ (" Δ_n is of order δ_n in probability") to mean that

$$\lim_{\lambda \rightarrow \infty} \limsup_{n \rightarrow \infty} P(|\Delta_n/\delta_n| > \lambda) = 0.$$

If in addition

$$\lim_{\epsilon \rightarrow 0} \limsup_{n \rightarrow \infty} P(|\Delta_n/\delta_n| > \epsilon) > 0,$$

we say that Δ_n is of precise order or precise size δ_n , in probability.

The “first term” in an Edgeworth expansion is the term of largest order that converges to zero as $n \rightarrow \infty$. The term in the expansion that does not depend on n is called the “0th term.”

Theorems are stated under explicit regularity conditions; *Propositions* are not given under explicit conditions.

Wherever possible we use the term *estimate*, rather than estimator, to indicate a function of the data that approximates an unknown quantity. For example, $\hat{\theta}$ is an estimate of θ . However, if θ is function-valued (e.g., if θ is a density) then we call $\hat{\theta}$ an estimator. It is sometimes difficult to be absolutely consistent in this notation.